SYMBOL LEGEND (IEC 60617 forms)

- Circuit breaker
- Circuit breaker, withdrawable
- Disconnector
- Switch-disconnector
- Fuse
- Earthing switch
- Current transformer (2 cores)
- Voltage transformer
- Surge arrester / SPD
- Busbar (weight per level)
- Transformer, 2 windings
- Earth
- Generator, 3-phase
- Rectifier (ac/dc), inverter (dc/ac)
- Static bypass switch
- Battery
- Adjustable inductor (Petersen coil)
- Resistor
- Busbar trunking / busway
- Energy meter (Wh) — Wh
- Protection relay, device no. — 51
- Optical arc sensor
- Interlock (mechanical / key)
- Protection / control signal
- Off-sheet connector

LINE CODE (black on white; no colour code)

130 kV conductor	20 kV conductor	0.4 / 0.69 kV conductor
130 kV busbar	20 kV busbar	0.4 / 0.69 kV busbar
UPS / critical busbar	protection / control	interlock

Voltage level is also stated on every busbar tag; genset and BESS circuits use the LV weight.

UPS COMPOUND SYMBOL (all sheets)

rectifier ac/dc (top), inverter dc/ac (bottom),
DC link with battery to the side;
drawn the same for IT UPS and mechanical UPS.

DRAWING LIST (Rev P2)

Drawing / sheet	Title
SLD-000	Legend, drawing list, notes
SLD-001 sh 1	130/20 kV substation
SLD-001 sh 2	20 kV main switchgear MSG-20
SLD-002 sh 1	Data hall DH-1: HV-DH1, blocks
SLD-002 sh 2	DH-1: UPS, busways, mechanical
SLD-003 sh 1	Power block A: T-A, G-A, MSB-A
SLD-003 sh 2	Power block A: UPS to rack
SLD-004	BESS plant 10 MW / 20 MWh

ABBREVIATIONS

ACB / VCB air / vacuum circuit breaker
ATS automatic transfer (open transition)
ARMS arc-reduction maintenance switch
BTS busbar trunking system
MSB / UOSB main / UPS output switchboard
SBS / MBB static / maintenance bypass
PCS / EMS power conversion / energy mgmt
ZSI zone-selective interlocking
LoM loss of mains (ROCOF / vector shift)
(A) assumed value

GENERAL NOTES

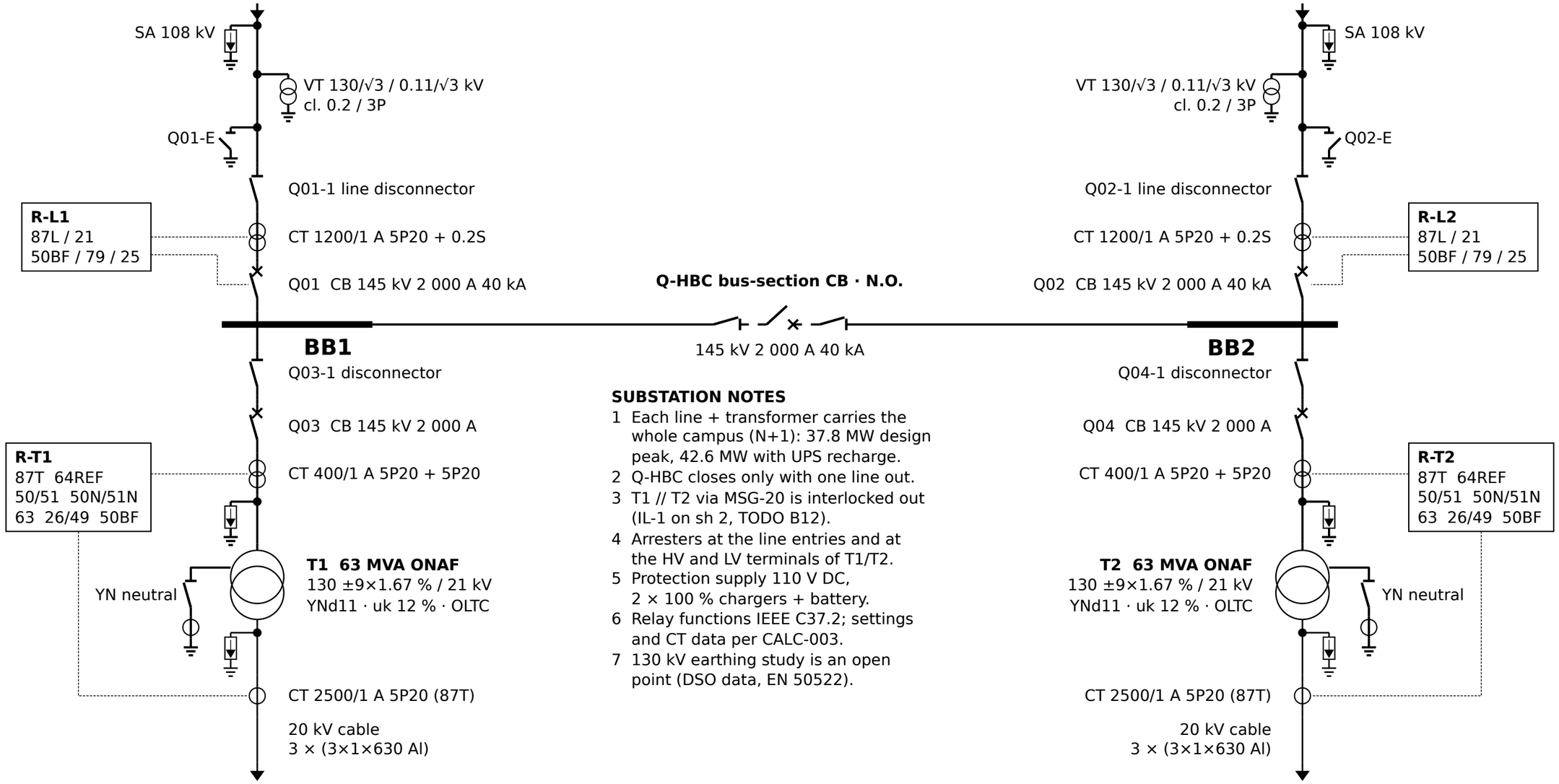
1. Concept drawings for a fictional campus; not for construction. Public standards, [ASSUMED] data.
2. Ratings per DB-001 Rev P2 and CALC-002/003/004.
3. Relay functions: IEEE C37.2 device numbers.
4. Minimum text 10 pt = 2.57 mm cap height at A3 (checked by scripts/check_drawings.py).

P2	2026-10-25	CALC-002/3/4 ratings, IEC symbols, ≥2.5 mm text
P1	2026-10-24	First concept issue

Nordic Hyperscale DC Campus (fictional) · 30 MW IT · SE3
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Legend, drawing list, notes
Symbols to IEC 60617 · lettering to ISO 3098
CONCEPT · NOT FOR CONSTRUCTION · scale NTS · A3

Drawing no.	Sheet	Rev
DWU-DC-E-SLD-000	**1 / 1**	**P2**



SUBSTATION NOTES

1 Each line + transformer carries the whole campus (N+1): 37.8 MW design peak, 42.6 MW with UPS recharge.
2 Q-HBC closes only with one line out.
3 T1 // T2 via MSG-20 is interlocked out (IL-1 on sh 2, TODO B12).
4 Arresters at the line entries and at the HV and LV terminals of T1/T2.
5 Protection supply 110 V DC, 2 × 100 % chargers + battery.
6 Relay functions IEEE C37.2; settings and CT data per CALC-003.
7 130 kV earthing study is an open point (DSO data, EN 50522).

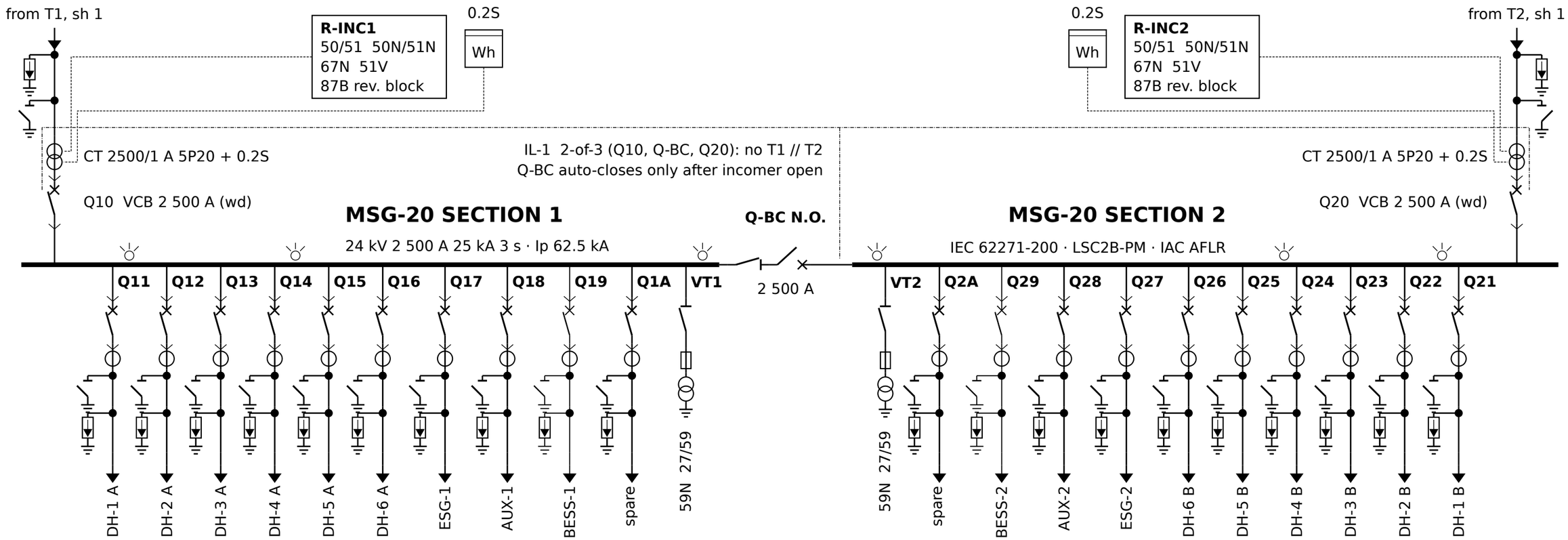
Short-circuit level (CALC-003/004)	Value
130 kV S''k max / min (A)	3 000 / 1 500 MVA
20 kV normal (S0)	13.0 kA, ip 34.4 kA
20 kV with BESS	13.3 kA
Worst abnormal + BESS	23.1 kA = 92 % of 25
T1 // T2 (not permitted)	ip 58.5 kA = 94 %

P2	2026-10-25	CALC-002/3/4 ratings, IEC symbols, ≥2.5 mm text
P1	2026-10-24	First concept issue

Nordic Hyperscale DC Campus (fictional) · 30 MW IT · SE3
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130/20 kV substation — sheet 1
H-arrangement, T1/T2 to the MSG-20 incomers
CONCEPT · NOT FOR CONSTRUCTION · scale NTS · A3

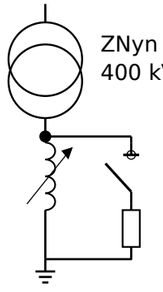
Drawing no.	Sheet	Rev
DWU-DC-E-SLD-001	**1 / 2**	**P2**



FEEDER SCHEDULE (relay settings per CALC-003)

Bay	Feeds	Device	CT	Protection / metering
Q10 / Q20	T1 / T2 incomer	VCB 2 500 A wd	2500/1 5P20 + 0.2S	51 SI 2 100 A TMS 0.16 50 6 000 A 0.65 s
Q11–16 / Q21–26	DH-1…6 feed A / B	VCB 630 A wd	400/1 5P20 15 VA	51 VI 320 A TMS 0.84 50 2 400 A 0.35 s
Q17 / Q27	ESG: ZNyn + coil	VCB 630 A wd	200/1 5P20	50/51, 51N, 59N
Q18 / Q28	Aux 500 kVA	VCB 630 A wd	100/1 5P20	50/51, 50N/51N
Q19 / Q29	BESS-1 / -2 5.5 MVA	CB 630 A wd	200/1 5P20 + 0.2S	see SLD-004
Q1A / Q2A	spare, equipped	VCB 630 A wd	400/1 5P20	as feeders
Q-BC	coupler (N.O.)	VCB 2 500 A wd	2500/1 5P20	50/51, IL-1

DETAIL B earthing bay Q17 / Q27



ZNyn earthing transformer
400 kVA aux winding (A)

Petersen coil ≥ 50 A, adjustable
10 … 60 A, auto-tuned
‖ 10 A resistor, switched
Ic 22 A/section, 44 A coupled,
+1 A BESS → 67N admittance

NOTES

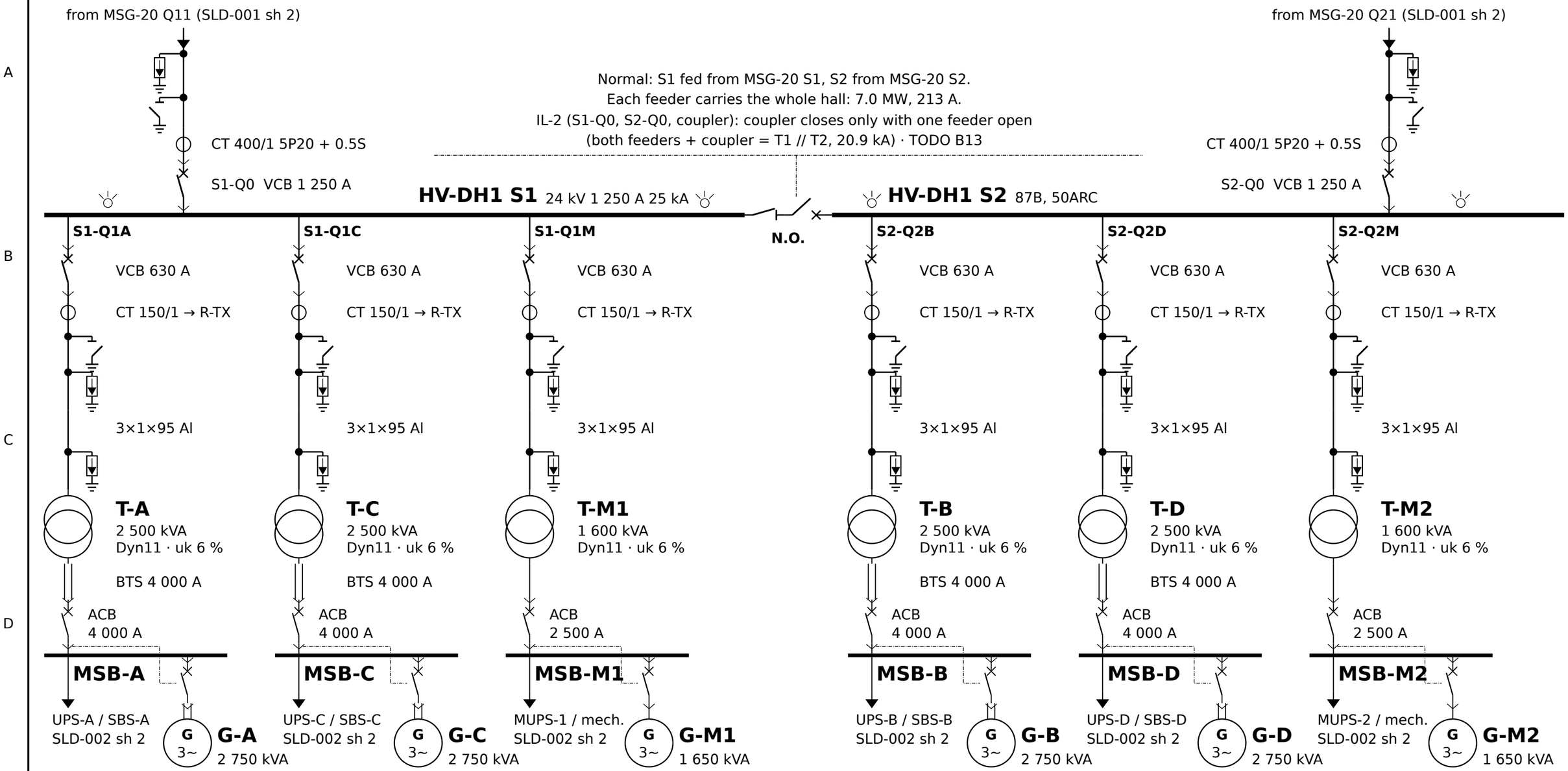
1 Resonant (compensated) earthing kept; 500 A low-resistance earthing rejected (CALC-003).
2 All feeders: 67N admittance earth-fault relays.
3 Reverse-blocking busbar protection + optical arc protection (50ARC, light + current) trips the section incomer and Q-BC (TODO B18).
4 Hall-board coupler closes only with one hall feeder open (IL-2, SLD-002 sh 1).
5 Every bay: SA 24 kV at the cable end, ES, EPMS 0.5S.
6 DH feeders A → HV-DH1 S1, B → S2 (SLD-002).
7 ESG-1/2: detail B. AUX: 500 kVA station aux.
8 BESS bays Q19 / Q29 → SLD-004 (TODO B22).

P2	2026-10-25	CALC-002/3/4 ratings, IEC symbols, ≥2.5 mm text
P1	2026-10-24	First concept issue

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20 kV main switchgear MSG-20 — sheet 2
2 sections, coupler, earthing bays, BESS bays
CONCEPT · NOT FOR CONSTRUCTION · scale NTS · A3

Drawing no.	Sheet	Rev
DWU-DC-E-SLD-001	**2 / 2**	**P2**



ATS (dash-dot): transformer ACB / genset ACB, open transition only (TODO B14). Gensets via BTS 4 000 A (B5).

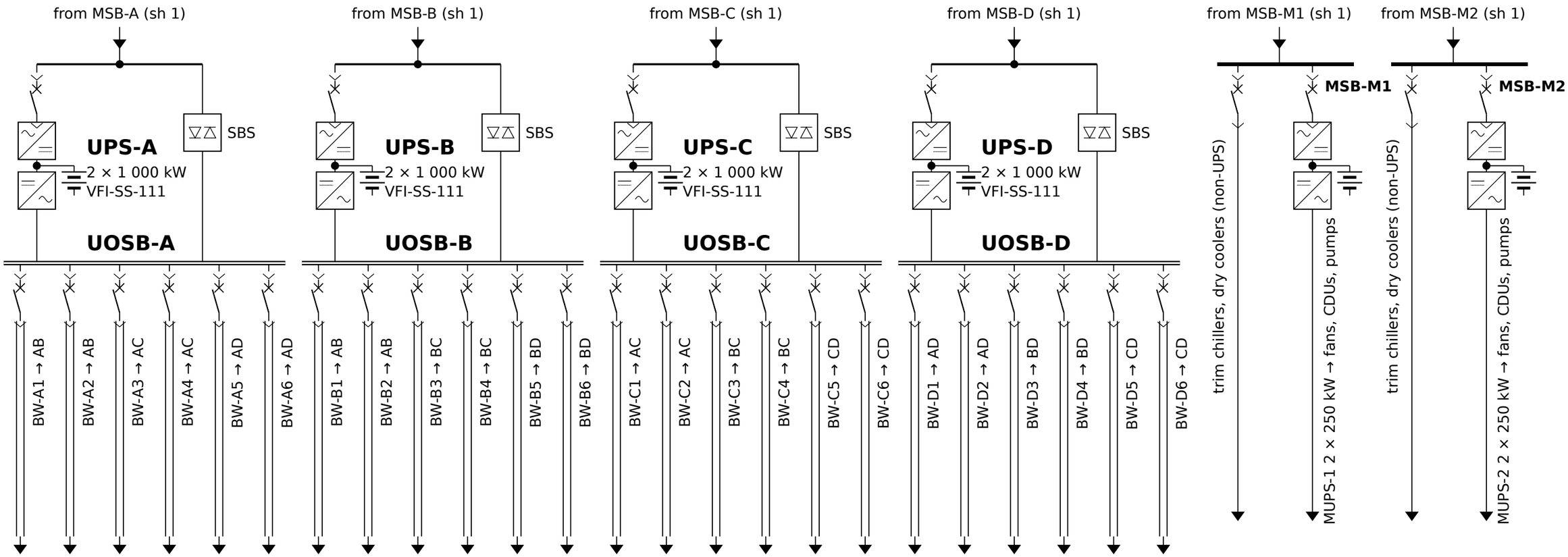
Item	Rating (Rev P2)	Check (CALC-002/3)
T-A…T-D	2 500 kVA 20/0.42 kV Dyn11, uk 6 %, Pk 19 kW	N-1 82 %
G-A…G-D	2 750 kVA / 2 200 kWe DCC, Isc ≥ 300 % 2 s	N-1 88 %
MSB-A…D	4 000 A, Icw 65 kA 1 s, Ipk 143 kA	52.3 kA
UPS-A…D	2 × 1 000 kW, 2 × 165 kWh LFP	N-1 83 %
UOSB-A…D	3 200 A, 65 kA 1 s	49.3 kA
T-M1/M2	1 600 kVA kept (1 250 kVA would be 83 %)	65 %
G-M / MUPS	1 650 kVA; mech. UPS 2 × 250 kW	step ≤ 33 %

P2	2026-10-25	CALC-002/3/4 ratings, IEC symbols, ≥2.5 mm text
P1	2026-10-24	First concept issue

Nordic Hyperscale DC Campus (fictional) · 30 MW IT · SE3
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Data hall DH-1 — sheet 1
HV-DH1, IT power blocks A–D (4N/3), mechanical 2N
CONCEPT · NOT FOR CONSTRUCTION · scale NTS · A3

Drawing no.	Sheet	Rev
DWU-DC-E-SLD-002	**1 / 2**	**P2**



4N/3 ALLOCATION

Pair-group	Busways (block 1)	Busways (block 2)	IT load
AB	BW-A1, BW-A2	BW-B1, BW-B2	0.83 MW
AC	BW-A3, BW-A4	BW-C1, BW-C2	0.83 MW
AD	BW-A5, BW-A6	BW-D1, BW-D2	0.83 MW
BC	BW-B3, BW-B4	BW-C3, BW-C4	0.83 MW
BD	BW-B5, BW-B6	BW-D3, BW-D4	0.83 MW
CD	BW-C5, BW-C6	BW-D5, BW-D6	0.83 MW

Each rack is dual-corded to the two blocks of one pair-group (2 busway runs per block).

LOADING (CALC-002)

Case	Block load	UPS	Busway
Normal	1.25 MW	63 %	307 A
Loss of 1 block	1.67 MW	83 %	614 A
DCIM limit	≤ 1.67 MW	—	Ir 800 A

NOTES

1. UOSB 3 200 A, Icw 65 kA 1 s; SPD type 2.
2. Busway MCCB 800 A current-limiting, Icu ≥ 50 kA, tested conditional rating ≥ 40 kA (IEC 61439-6).
3. Busway 800 A, Icw 36 kA; tap-off 63 A MCCB Icu ≥ 36 kA; rack PDU SCCR ≥ 15 kA (CALC-003).
4. Battery per UPS module LFP 165 kWh (5 min).
5. Block A detail: SLD-003 sh 1–2.

P2	2026-10-25	CALC-002/3/4 ratings, IEC symbols, ≥2.5 mm text
P1	2026-10-24	First concept issue

Nordic Hyperscale DC Campus (fictional) · 30 MW IT · SE3
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Data hall DH-1 — sheet 2
UPS, UOSB, busways to 4N/3 pair-groups; mechanical 2N
CONCEPT · NOT FOR CONSTRUCTION · scale NTS · A3

Drawing no.	Sheet	Rev
DWU-DC-E-SLD-002	**2 / 2**	**P2**

from HV-DH1 S1-Q1A (SLD-002 sh 1)

S1-Q1A trip
VCB 630 A 25 kA
CT 150/1 5P20 10 VA
Ith ≥ 25 kA 1 s

R-TX
50/51 Is 100 A, IEC EI, TMS 1.0
50 I≫ 1 500 A, 0.05 s
50N/51N · 86 lock-out

3×1×95 Al 12/20 kV

SA 24 kV

T-A
2 500 kVA 20/0.42 kV
Dyn11 · uk 6 % · cast resin
Pk 19 kW · P0 3.1 kW

BTS 4 000 A ≈ 10 m

Wh
CT 4000/1 → Wh

Q-UI ACB 4 000 A
Ir 3 600 A · ARMS

IL-3 ATS: Q-UI / Q-GI open transition only

G-A
G 3~
2 750 kVA / 2 200 kWe DCC
Isc ≥ 300 % for 2 s
HVO100, 24 h

CT 4000/1 → 51V, Wh

BTS 4 000 A ≈ 40 m

R-ARC
50ARC light + I
trip Q-UI, Q-GI
intertrip S1-Q1A

Q-GI ACB 4 000 A
Isd ≤ 2.5 × Ir

I″k 52.3 kA / ip 122 kA

MSB-A
4 000 A · Icw 65 kA 1 s · Ipk ≥ 143 kA

Q-R1 2 000 A	**Q-R2** 2 000 A	**Q-SB** 3 200 A	**Q-MB** 3 200 A	**Q-LB** 2 500 A	**Q-AUX** 250 A	SPD T1+2
UPS-A1 rectifier (sh 2)	UPS-A2 rectifier (sh 2)	BTS 3 200 A ≈ 25 m → SBS-A (sh 2)	MBB-A (sh 2)	load-bank connection	block aux. lighting, BMS	

LV TRIP SETTINGS (CALC-003)

Device	Ir	tr	Isd	tsd	Other
Q-UI incomer	3 600 A	3 s	4 × Ir	0.3 s	ZSI, ARMS
Q-SB (SBS)	2 900 A	1 s	2.5 × Ir	0.2 s	ZSI
Q-GI genset	4 000 A	12 s	2.5 × Ir	0.4 s	—
Busway MCCB	800 A	—	4 × Ir	0.1 s	Ii 12 × Ir

NOTES
1. ATS: break-before-make, no closed transition (closed: 81.8 kA / ip 187 kA > MSB rating).
2. ZSI Q-UI ↔ outgoing ACBs; ARMS on Q-UI.
3. Arc energy MSB line side 96 → 3.2 cal/cm², bus 16.8 → 2.4–4.8 cal/cm² (IEEE 1584).
4. Ratings are [ASSUMED] concept values.

P2	2026-10-25	CALC-002/3/4 ratings, IEC symbols, ≥2.5 mm text
P1	2026-10-24	First concept issue

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Power block A — sheet 1
20 kV feeder, T-A, genset G-A, MSB-A (typical of 24)
CONCEPT · NOT FOR CONSTRUCTION · scale NTS · A3

Drawing no.	Sheet	Rev
DWU-DC-E-SLD-003	**1 / 2**	**P2**

from MSB-A Q-R1 (sh 1)

from MSB-A Q-R2 (sh 1)

BTS 3 200 A from Q-SB (sh 1)

from MSB-A Q-MB (sh 1)

UPS-A1
1 000 kW · VFI-SS-111
LFP 165 kWh (5 min EoL)

Q-O1
ACB 1 600 A

UPS-A2
1 000 kW · VFI-SS-111
LFP 165 kWh (5 min EoL)

Q-O2
ACB 1 600 A

SBS-A
3 200 A static bypass

Q-SBO
ACB 3 200 A

Q-MBB
ACB 3 200 A
maint. bypass

UPS output bus 3 200 A

Q-UO
ACB 3 200 A

IL-4 key interlock: Q-UO / Q-MBB
(one key, never both closed)

UOSB-A 3 200 A · Icw 65 kA 1 s

49.3 kA

MCCB BW-A1
800 A, Icu ≥ 50 kA
current-limiting

busway 800 A, Icw 36 kA
cond. ≥ 40 kA (IEC 61439-6)

37.1 kA

MCCB BW-A2
busway 800 A
as BW-A1

MCCB BW-A3
busway 800 A
as BW-A1

MCCB BW-A4
busway 800 A
as BW-A1

MCCB BW-A5
busway 800 A
as BW-A1

MCCB BW-A6
busway 800 A
as BW-A1

SPD T2

tap-off 63 A MCCB
Icu ≥ 36 kA
25.1 kA

B-cord from BW-B1 (block B)

PDU-A

PDU-B

14.0 kA · PDU SCCR ≥ 15 kA

rack: 2 PSUs, each 100 %

EQUIPMENT SCHEDULE

Item	Rating (Rev P2)	Basis
UPS-A1/A2	1 000 kW each, VFI-SS-111	N-1 83 %
Battery	LFP 165 kWh per module, ≤ 8 C EoL	B1
Rectifier breaker	Q-R1/Q-R2 2 000 A (1 660 A max)	B2
SBS / MBB	3 200 A, BTS 3 200 A from MSB	B4
Busway MCCB	800 A, Icu ≥ 50 kA, cond. ≥ 40 kA	B3, B15
Tap-off / PDU	63 A Icu ≥ 36 kA; SCCR ≥ 15 kA	B16

NOTES
1. Fault levels in boxes: I″k per CALC-003.
2. SBS is centralised per block; MBB is external wrap-around (IL-4).
3. A2…A6 identical to BW-A1; pairing per SLD-002 sh 2.

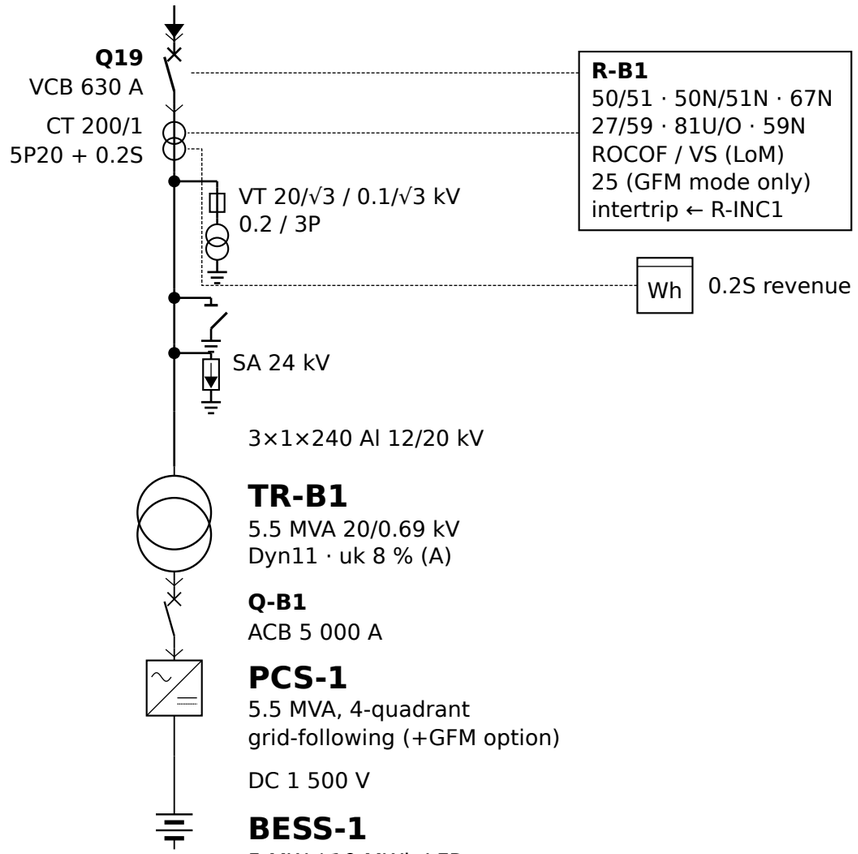
P2	2026-10-25	CALC-002/3/4 ratings, IEC symbols, ≥2.5 mm text
P1	2026-10-24	First concept issue

Nordic Hyperscale DC Campus (fictional) · 30 MW IT · SE3
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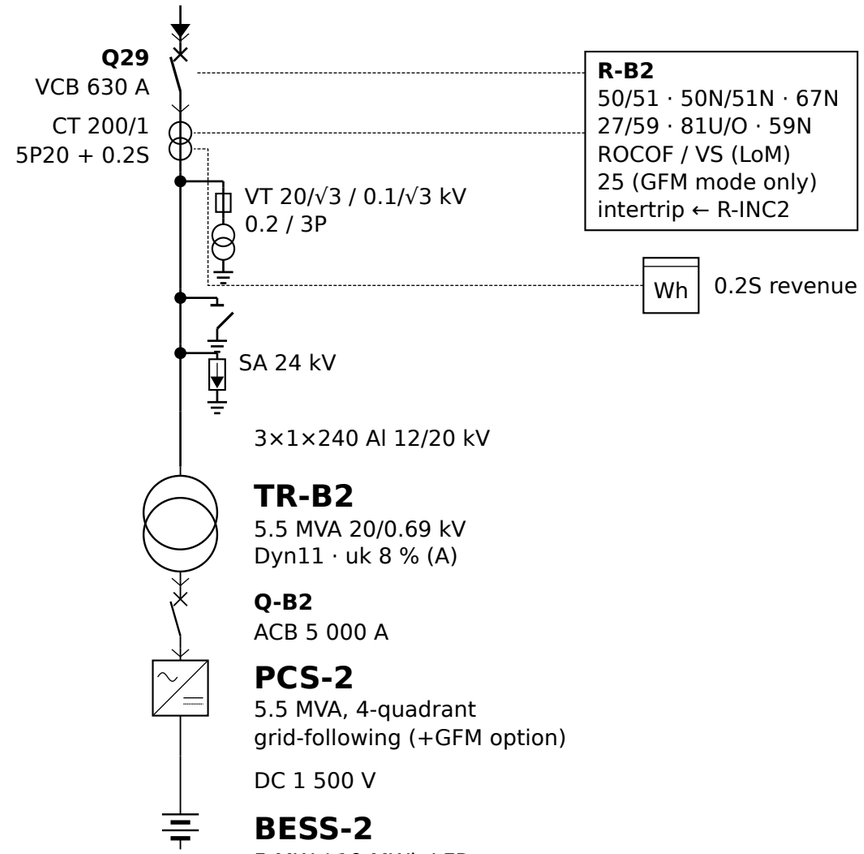
Power block A — sheet 2
UPS-A1/A2, SBS, MBB, UOSB-A, busway to rack (typical)
CONCEPT · NOT FOR CONSTRUCTION · scale NTS · A3

Drawing no.	Sheet	Rev
DWU-DC-E-SLD-003	**2 / 2**	**P2**

from MSG-20 Q19 (SLD-001 sh 2)



from MSG-20 Q29 (SLD-001 sh 2)



EMS / PPC priorities

1. Import ≤ 40 MW incl. recharge
2. SOC reserve before peaks
3. FCR down headroom
4. Markets (FCR, mFRR)

Fallback: IT power capping → R-B1, R-B2, PCS-1/2

BESS SCHEDULE

Item	Rating (Rev P2)	Note
Bays Q19 / Q29	VCB 630 A, CT 200/1 5P20 + 0.2S	unit 159 A at 20 kV
TR-B1 / TR-B2	5.5 MVA 20/0.69 kV Dyn	delta at 20 kV
PCS-1 / PCS-2	5.5 MVA each, 11 MVA total	p.f. ≥ 0.95 at PCC
BESS-1 / BESS-2	5 MW / 10 MWh LFP each	10 MW / 20 MWh
I″k contribution	≤ 1.2 × In (PCS limit), 13.0 → 13.3 kA	CALC-004

NOTES

1. Not counted as Tier III / EN 50600 Class 3 capacity; firm IT gain +0.84 MW (peak shaving +2.4 MW).
2. No island operation in the base design: 20 kV relays do not clear an inverter-fed island fault (381 A, R-FDR ≈ 60 s).
3. Containers ≥ 10 m (A) from halls, transformers, fuel; UL 9540A, NFPA 855, IEC 62933-5-2.
4. Harmonic study (IEC/TR 61000-3-6) open.

P2	2026-10-25	CALC-002/3/4 ratings, IEC symbols, ≥2.5 mm text
P1	2026-10-24	First concept issue

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BESS plant — 10 MW / 20 MWh
2 units, MSG-20 bays Q19 / Q29, protection and EMS
CONCEPT · NOT FOR CONSTRUCTION · scale NTS · A3

Drawing no.	Sheet	Rev
DWU-DC-E-SLD-004	**1 / 1**	**P2**